2. (i) True (ii) False (iii) False (iv) True (v) False

3. (i) $\frac{1}{8}$ (ii) 32 (iii) $\frac{-216}{125}$ (iv) 16 (v) 6

4. (i) $\frac{64}{15625}$ (ii) $\frac{4}{5}$ (iii) 1024 5. (i) $\frac{21}{2}$ (ii) 5 (iii) 1

6. (i) 1 (ii) $\frac{3}{2}$ (iii) 72 7. (i) $x = 5$ (ii) $x = 6$

8. (i) $6 \times 10^3 + 5 \times 10^1 + 4 \times 10^0 + 3 \times 10^{-1} + 2 \times 10^{-2} + 1 \times 10^{-3}$
(ii) $8 \times 10^2 + 9 \times 10^1 + 7 \times 10^0 + 1 \times 10^{-1} + 4 \times 10^{-2}$

9. (i) 87652.0407 (ii) 5050.505 (iii) 0.000000000025

10. (i) 4.678×10^{11} (ii) 1.972×10^{-6} (iii) 1.642398×10^3
(iv) 1.083×10^{12} cu. km (v) 1.6×10^{-24}

11. (B) $\frac{-2}{5}$ 12. (A) $\frac{-1}{32}$ 13. (D) $-\left(\frac{1}{4}\right)^2 = 16^{-1}$ 14. (C) 6 15. (D) 2.02×10^{-10}

Exercise 1.7

Miscellaneous Practice Problems

1. 4 kg 300 gm 2. $10\frac{2}{5}$ litre 3. both are correct 4. 6 m 5. 552 cm^2
6. 9 cm and 10 cm 7. 15 decimetre 8. 625 9. 8
10. (i) 4.8×10^3 (ii) 1.152×10^5 (iii) 4.2048×10^7 (iv) 4.2048×10^9

Challenging Problems

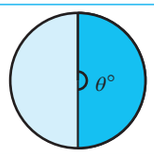
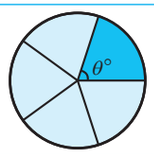
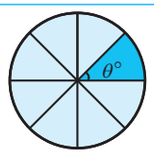
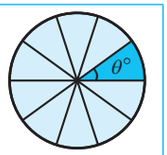
11. 900 km 13. 320 gm 14. $\frac{9}{20}$ 15. $x = 3$ 16. $\frac{12}{11}$ 17. No, 64
18. 58.85 19. 7.979×10^5 20. 8^{100}, 2^{600}, 3^{500}, 4^{400}, 16^{25}

2. MEASUREMENTS

Exercise 2.1

1. (*i*) π (ii) chord (iii) diameter (iv) 12 *cm* (v) circular arc
2. (i) c (ii) d (iii) e (iv) b (v) a

3.

Sectors				
Central angle of each sector (θ°)	180°	72°	45°	36°

4.

	Length of arc (approx.)	Area of sector (approx.)	Perimeter of sector (approx.)
(i)	12.56 *cm*	100.48 cm^2	44.56 *cm*
(ii)	13.19 *cm*	41.54 cm^2	25.79 *cm*

5. (i) 240 m^2 (ii) 337.5 cm^2 6. (*i*) $\theta = 120°$ (*ii*) $\theta = 72°$ 7. $30\pi\, m$
8. $980\pi\, cm^2$ 9. 706.5 cm^2 (approximately) 10. 1232 cm^2 (approximately)

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